The universal property of set-quotients: maps out of A/R are the functions on A constant on R-related pairs

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:quotient-ump' (source lines 1222-1225, Lemma 6.10.3); the proof environment begins on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence, which identifies the universal property being proved with that of a set-coequalizer.

Editorial note. The retained counter values (chapter 6, section 10, theorem offset 2) make the retained lemma print as Lemma 6.10.3, matching the original book numbering.

Editorial note. The single external reference points at the surjectivity of q, which the retained proof invokes by label; the counted proof is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: the universal property of set-quotients (Lemma 6.10.3).

Allowed standard prerequisites (editorial).

- Recursion principle of the set-quotient. For a set B and a map f : A -> B that respects R, there is fbar : A/R -> B with fbar(q(a)) := f(a).. It defines the quasi-inverse of precomposition with q at source lines 1227-1229, and its computation rule is exactly the right-inverse statement at line 1230.
- Surjectivity of the quotient map. For every x : A/R there merely exists a : A with q(a) = x.. It supplies the point a at source line 1233 that lets the left-inverse computation begin.
- Mere propositions and the truncation of the goal. If the goal type is a mere proposition then a merely existentially quantified hypothesis may be used purely, i.e. one may assume an actual witness.. It is what turns the merely existing a at source line 1233 into a usable witness at 1234.

Source reference. Original lemma 6.10.2: sources/hott-hits/pinned-original-source.tex, source lines 1210–1210. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

We can now prove that the set-quotient has the expected universal property of a (set-)coequalizer.

Lemma 6.10.3. *For any set B, precomposing with q yields an equivalence*

$$(A/R \to B) \simeq \Big(\sum_{(f:A\to B)} \prod_{(a,b:A)} R(a,b) \to (f(a) = f(b)) \Big).$$

Proof. The quasi-inverse of $- \circ q$, going from right to left, is just the recursion principle for A/R. That is, given $f : A \to B$ such that $\prod_{(a,b:A)} R(a,b) \to (f(a) = f(b))$, we define $\bar{f} : A/R \to B$ by $\bar{f}(q(a)) :\equiv f(a)$. This defining equation says precisely that $(f \mapsto \bar{f})$ is a right inverse to $(- \circ q)$.

For it to also be a left inverse, we must show that for any $g : A/R \to B$ and $x : A/R$ we have $g(x) = \overline{g \circ q}(x)$. However, by original lemma 6.10.2 there merely exists a such that $q(a) = x$. Since our desired equality is a mere proposition, we may assume there purely exists such an a, in which case $g(x) = g(q(a)) = \overline{g \circ q}(q(a)) = \overline{g \circ q}(x)$. □